Customer Lifetime Value Optimization Using Machine Learning

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Data-driven retention targeting nearly doubles ROI under fixed marketing budgets.

Background / Problem

- Churn drives revenue loss; campaigns miss high-value customers
- Predict churn probability
- Estimate Expected Loss (P(churn) × CLV)
- Optimize spend under budget
- Compare targeted vs. random strategies

Results

ROC-AUC: 0.851
Revenue Saved: $224,684
Efficiency: 1.94×

Simulation (25*Kbudget*, 50/customer, 15% reduction):
Targeted: $224,684 saved | ROI 799%
Random: $116,083 saved | ROI 364%

- Sensitivity: robust across 5–25% churn, 25–100 cost

Analysis / Conclusions

- Rank by Expected Loss → maximize revenue saved under budget
- Sensitivity + budget scaling: results robust

Business Impact
- ~2× revenue improvement vs random
- Scalable across budget levels
- Reproducible ML + simulation pipeline

Top drivers: tenure, monthly charges, contract.

Methods

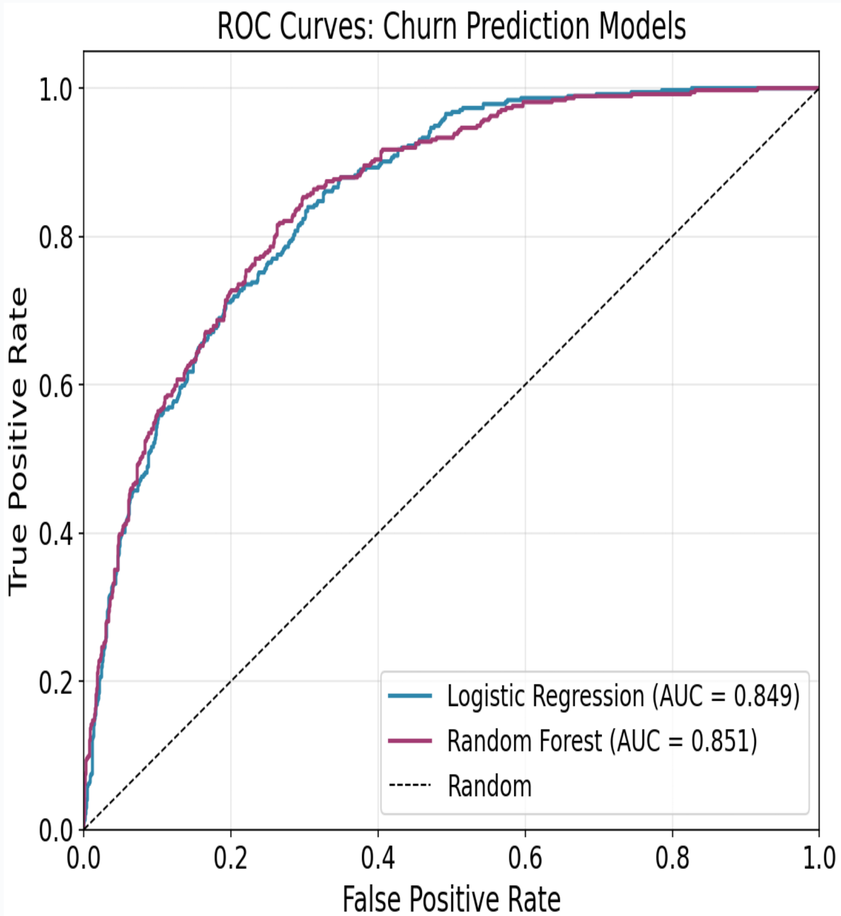
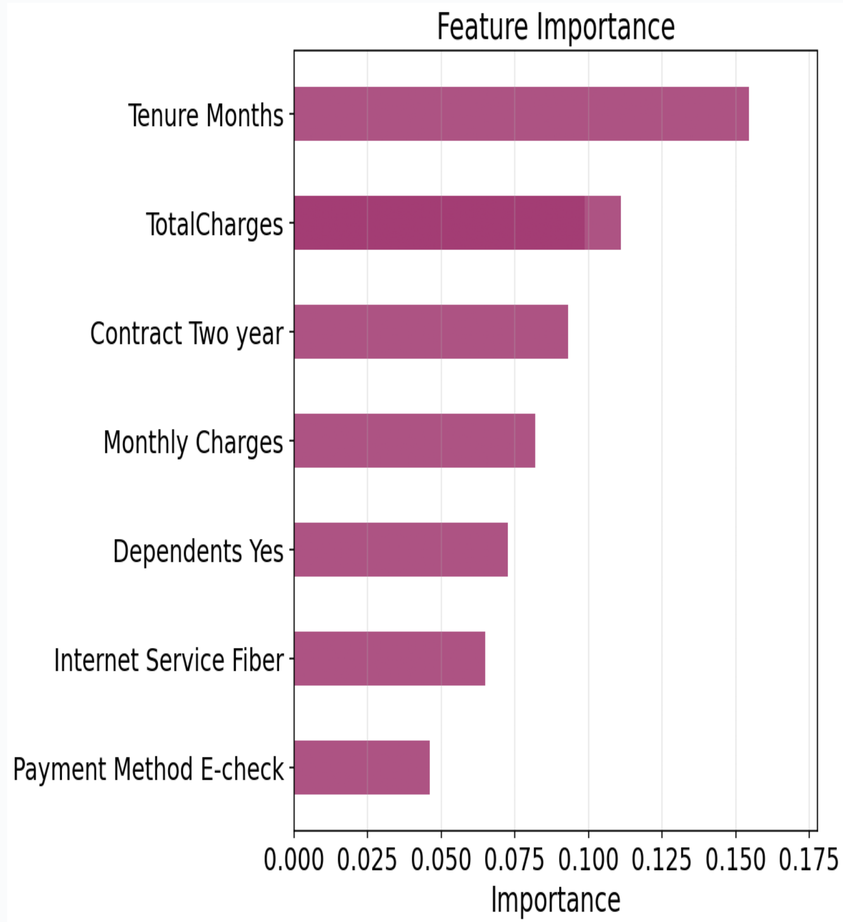
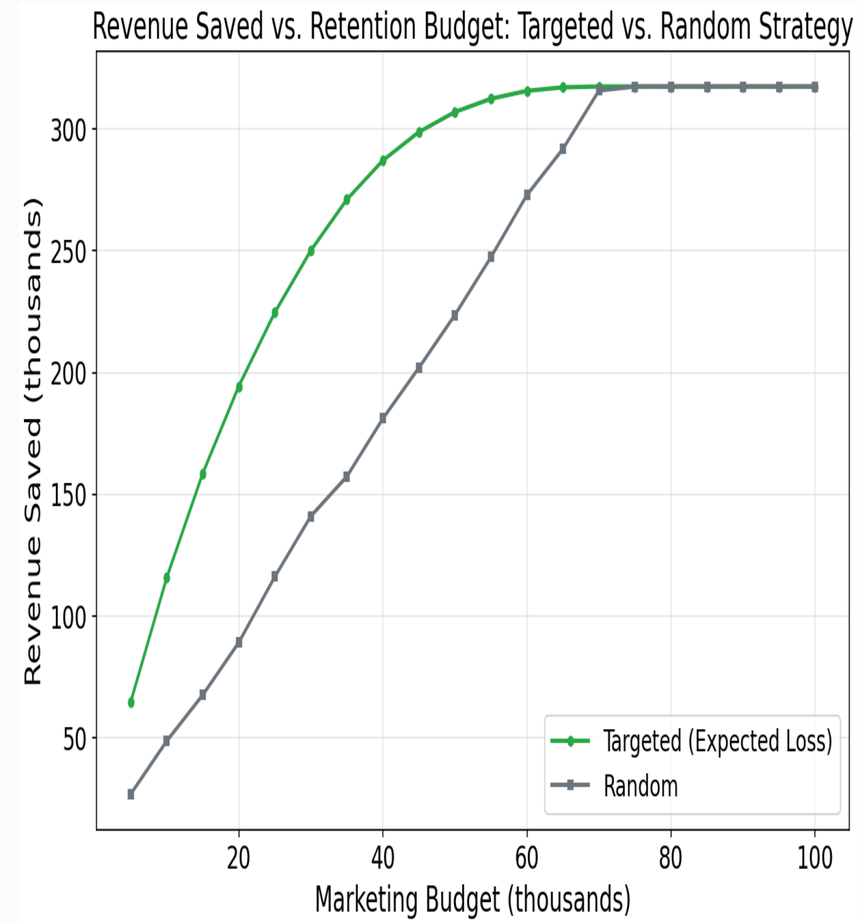
Data
- IBM Telco: 7,043 customers, 31 features
- Preprocessing: TotalCharges, Churn mapping, one-hot encode

Modeling
- Logistic Regression + Random Forest
- Metrics: ROC-AUC, precision, recall

Simulation
- Expected Loss = P(churn) × CLV
- Targeted vs. random under fixed budget
- Revenue saved, ROI, targeting efficiency

References

- IBM Telco Churn Dataset
- IBM: github.com/IBM/telco-customer-churn-on-icp4d
- Kaggle: kaggle.com/datasets/yeanzc/telco-customer-churn-ibm-dataset
- Apache-2.0 | 7,043 customers, 31 features

GitHub

- **Repository:**
github.com/Subhadra-Mishra-iub/customer-lifetime-value-optimization
- **Dataset:**
kaggle.com/datasets/yeanzc/telco-customer-churn-ibm-dataset
- **Description:**
Churn prediction and retention budget optimization pipeline.